Formalizing AI Scientist Systems: A Component Framework and Theoretical Analysis

Jaesol Shin
WeDataLab
ysys143@gmail.com
https://github.com/ysys143

Post-submission revised manuscript — not the version of record. This manuscript was refined after the CAISc 2026 submission deadline and is *not* the paper that was reviewed and archived. The version of record is the camera-ready deposited on OpenReview; nothing here alters it. The detailed revision chronology is retained with its rationale in the repository release documentation. For bibliographic identity, cite the version of record; when relying on a claim corrected here, cite or identify this dated revision alongside it and do not attribute the corrected claim to the archival text. The correction should also be linked from the official record when that repository permits.

Revised: 2026-07-19; last revised 2026-07-22.

Abstract

Existing surveys organize AI Scientist systems by capabilities and workflow stages. We propose Servo (S, G, E, V, M, π), which operationalizes the sequential information structure of POMDPs and the experiment-selection logic of BED as diagnostic distinctions for source-bounded analyses of AI Scientist systems. The six-component core comprises the **S**earch space, hypothesis **G**enerator, **E**xperiment executor, **V**alidator, **M**emory, and search **P**olicy (π); the environment observation law O_{env} is an explicit conceptual boundary, while artifact state A and the optional artifact synthesis or revision interface W_A are recorded boundary objects rather than additional core components. We apply the schema to six versioned cases from five lineages and seven selected, frozen domain anchors. These distinctions separate observation from evaluation and distinguish outcome-conditioned repair or score improvement from hypothesis-discriminating experimentation and from the rules determining which candidates are executed. As an auxiliary corpus-level observation, none of the bounded sources reports an explicit posterior-based or expected-information-gain criterion for experiment selection. The case and anchor records illustrate use of the vocabulary; they are not a representative sample, prevalence estimate, causal test, or demonstration that Servo is superior to another framework. Appendix A uses standard Bayesian results only to delimit observation, validation, and novelty in fixed idealized settings. We identify three open problems as prospective design questions: independently validated automated novelty assessment, integration of information-seeking policies, and multi-fidelity integration.

1 Introduction

AI Scientist systems—agents that generate hypotheses, execute or coordinate experiments, and synthesize knowledge with varying human authority—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Feng et al., 2026, Liu et al., 2026]. Capability- and stage-based surveys provide shared organizing frameworks [Wei et al., 2025, Tie et al., 2025], but different analytical purposes foreground different system boundaries. The questions pursued here are narrower: *Which evaluation event is connected to which later action? Which artifact version carries that connection? Which kind of feedback, if any, does a bounded reported configuration support, and does the source resolve* *it at the level of an individual occurrence? What evidence would support novelty assessment or information-seeking experimental policy?*

We propose Servo[1] (S, G, E, V, M, π), which operationalizes the sequential structure of POMDPs and the experiment-selection logic of Bayesian experimental design as diagnostic distinctions for source-bounded analyses of AI Scientist systems (Section 4). The contribution is this set of distinctions together with its source-traceable application to six versioned cases from five lineages: applied to these cases, the distinctions decompose systems commonly described using terms such as iterative or closed-loop into distinct feedback structures—separating an observation from its subsequent evaluation, and distinguishing the information on which follow-up selection depends, the objective that selection serves, and the rule determining which candidates are executed. Seven selected source anchors are described separately (Section 6).

Scope. Servo is a non-exhaustive system-level diagnostic schema. Each case is a versioned system configuration in a bounded task regime, identified by case, lineage, version, configuration, and task-regime identifiers. Inclusion records source-traceable variation in execution, evaluation, artifacts, routing, policy, authority, or fidelity; it is not a claim that the cases exhaust or represent AI Scientist systems.

2 Background

Following Kaelbling et al. [1998], we borrow the organizing idea that a policy maps information available under incomplete observation to an action. Servo denotes the source-reported decision-time information by I_t. It may comprise an observation history, retrieved memory, validator outputs, or a reported controller, predictive, or recurrent state. A fully instantiated POMDP would require a world-state space $\mathcal{X}$, action space $\mathcal{A}$, transition kernel, observation kernel, reward function, initial belief, and discount. Servo does not specify those objects and is not a POMDP instantiation. Its S is a candidate search space rather than the world state. Servo does not infer an unreported belief state; only when a bounded source defines an explicit belief or posterior do we use the specialization $I_t = b_t$. Appendix A separately analyzes fixed statistical models. Posterior concentration is not any of Servo's typed feedback relations, and no relation's evidence status defines a trustworthy scientific outcome. An outcome study must predeclare a gate-independent endpoint Y_q for each target property q. Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides one possible normative information criterion for π: select d^* maximizing Expected Information Gain (EIG),

$$d^* = \arg\max_{d \in \mathcal{D}} \ \mathrm{EIG}(d) = \arg\max_{d} \ \mathbb{E}\left[H(\theta) - H(\theta \mid y, d)\right] \tag{1}$$

where $H(\theta)$ is prior entropy over parameters θ and $H(\theta \mid y, d)$ is expected posterior entropy after observing y under d. EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022]. Robot Scientist [King et al., 2004] and GNoME [Merchant et al., 2023] illustrate domain-specific active-learning policies; the bounded LLM case sources used here do not state an EIG objective.

3 Related Work

AI Scientist systems and direct survey frameworks. Individual systems have been characterized in detail [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023], while recent surveys provide shared comparison frameworks. Wei et al. [2025] integrate capabilities and workflow, and Tie et al. [2025] organize systems by methodological stages and abstraction layers. Servo instead foregrounds versioned case identity, events, artifacts, routes, and typed functional relations with per-relation evidence. This is a difference in analytical unit, not evidence that Servo is more complete or useful.

Autonomous-science frameworks. Automated feedback architectures predate the LLM era: the Robot Scientist [Sparkes et al., 2010] integrated hypothesis generation, robotic experimentation,

[1]The servomechanism metaphor motivates tracing a feedback path; it does not imply that every recorded path is reliable, epistemic, autonomous, or scientifically successful.

statistical testing, model update, and active experiment selection [King et al., 2004]. More recent architectures include InternAgent (formerly NovelSeek) [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025] and Co-Scientist [Gottweis et al., 2026]. We therefore position Servo not as the first unified framework but as a vocabulary that conceptually distinguishes observation generation from evaluation while operationally recording documented execution-anchored evidence routes, artifacts, and acceptance. Its diagnostic value is illustrated, not independently tested against another framework.

Decision-theoretic foundations. The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical, with a long history of Bayesian-optimal experiment selection in automated science—most directly the Robot Scientist's active-learning policy [King et al., 2004]. Servo does not claim a complete POMDP or priority for BED; as Section 4 sets out, it operationalizes their sequential-information and experiment-selection structure as diagnostic distinctions. Its machine-readable contract records validation target, evidence, evaluator, decision role, artifact state, and routed edge without treating those fields as a reliability hierarchy.

Formal models and artifact infrastructure. Taniguchi et al. [2025] model collective scientific inference, whereas Servo records one system-level case and deliberately compresses social organization. Liu et al. [2026] foreground artifact format, provenance, and verification. The Servo event–artifact schema can record such artifacts as versioned A states and, where the source reports it, an artifact synthesis or revision interface W_A; this is a boundary mapping rather than subsumption.

Provenance and workflow-run standards. PROV-DM already defines entities, activities, agents, generation, usage, derivation, revision, and association [Moreau and Missier, 2013]. Workflow Run RO-Crate similarly records a workflow plan and run with code and input/output artifacts [Workflow Run RO-Crate Community, 2024]. Servo does not redefine these base provenance objects. Its artifacts map to PROV entities, events to activities or run actions, producers and consumers to generation and usage, and human or system actors to agents and associations. The added profile is narrower: scientific target properties, validator-reliability records, typed epistemic and control routes, bounded case identity, component-specific authority, and typed functional relations with per-relation evidence claims. Table 1 states this reuse boundary; formal conformance would additionally require an executable serializer and the relevant conformance tests.

Servo record	Existing analogue	Servo-specific scientific semantics
Artifact/version	PROV Entity; derivation/revision	Version identity required by a bounded functional-relation claim
Event	PROV Activity; RO-Crate run action	Target property, evaluator, reliability evidence, update semantics
Actor/route	Agent/Association; generation/usage	Component authority and typed epistemic, control, revision, or mediation role
Functional relation/claim	Connected provenance subgraph	Typed source–target relation with its per-relation evidence status

Table 1: Construct-level provenance crosswalk. Servo is a scientific-diagnostic profile over reusable provenance concepts, not a replacement base ontology.

Framework	Unit and aim	Foregrounded resolution	Deliberate omission / tradeoff	Source
Servo	System-level discovery loop comparison	Components; property-specific validator channels; operational versus terminal feedback	Compresses agent communication and social organization; not a complete generative model	This work
Wei et al.	Capability–workflow–domain synthesis	Five foundational capabilities and four dynamic discovery stages	Validator-channel identity and operational versus terminal routing are not the organizing unit	Wei et al. [2025]
Tie et al.	Six-stage methodology by four abstraction layers	Stage coverage across domains, tasks, architectures, and capabilities for 50+ systems	Does not separately encode memory, policy, or complete validator feedback paths	Tie et al. [2025]

Table 2: Direct construct-level comparison with AI Scientist survey frameworks, using each framework's native unit. This is not a performance ranking. Servo's claimed increment is selected system-level channel and interface resolution, not the first unified framework or information that the alternatives are incapable of representing.

POMDP, InternAgent, CPC-MS, and ARA remain complementary rather than direct survey-taxonomy baselines. POMDP supplies a sequential decision model when states and kernels are specified [Kaelbling et al., 1998]; InternAgent foregrounds implemented multi-agent organization [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025]; CPC-MS models community-level epistemic dynamics [Taniguchi et al., 2025]; and ARA foregrounds artifact provenance and replay [Liu et al., 2026]. Servo does not subsume these analytical units.

4 The Servo Framework

Servo does not merely borrow POMDP and BED terminology, nor does it score whether a system implements either: these are constitutive analytical resources from which its diagnostic categories are derived, while a complete POMDP instantiation or a formal BED objective is a special case rather than a requirement. From the POMDP, Servo takes the sequential relation among action, observation, and decision information; it then distinguishes an execution- or environment-generated observation from the system's subsequent assessment of that observation. From BED, it takes experiment selection under uncertainty as a normative reference and separates the information used for selection, the objective served, and the rule determining which candidates are executed.

We define an AI Scientist system as a tuple (S, G, E, V, M, π) with the following components:

$$\text{Servo} = (S,\ G,\ E,\ V,\ M,\ \pi) \tag{2}$$

S**: Search Space** The currently represented candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. S_t may be fixed or expanded; calling a process open-ended requires an explicit expansion rather than merely searching a large fixed set.

$G : M \times S \to (\mathcal{C}, \Delta S)$**: Hypothesis Generator** A function that proposes both newly instantiated candidates and an optional search-space expansion. We set $S_t^+ = \text{expand}(S_t, \Delta S_t)$ and require $\mathcal{C}_t \subseteq S_t^+$. Fixed-space search has $\Delta S_t = \varnothing$; progressive widening or a newly introduced representation must be recorded in ΔS_t. G supplies options and expansions but does not select the next system action.

$E \equiv E_{\text{exec}} : h \times d \times P \times f \to u$**: Experiment Executor** The system-controlled component that turns a selected hypothesis, design, protocol, and fidelity into an executed intervention or computational action u. It may be a code runner, simulator, robot, or protocol orchestrator, with fidelity ranging from computational proxy to physical wet-lab execution. The stochastic response of nature or a simulator is not the executor itself. We denote that auxiliary environment and measurement law by $O_{\text{env}}(\mathrm{d}o \mid x, u)$, where x is an environment state rather than a Servo component. Hence an observation is drawn as $o \sim O_{\text{env}}(\cdot \mid x, E_{\text{exec}}(h, d, P, f))$, with a Dirac law as the deterministic special case. O_{env} is not a seventh Servo component or a field inferred by the architectural coding; an explicit statistical model is required before a Bayesian specialization $I_t = b_t$, belief update, or EIG is computable. V remains a separate contextual assessment of the realized evidence.

$V = \{V_j : (z_j, A_v) \to r_j\}_{j \in \mathcal{J}}$**: Validator events** A family of context-dependent assessment functions rather than one compulsory post-observation call. A_v identifies every versioned artifact consumed by the event; its producer, version, and case boundary must resolve in the artifact records. The contextual input $z_j = (h, d, P, o, M, K_{\text{ext}}, \tau)$ may additionally contain the candidate claim or hypothesis, design, protocol, observation, internal memory, an external reference corpus, and an evaluation-time cutoff. Its result r_j may contain property-specific components for executability, empirical adequacy, reproducibility, novelty, significance, and acceptance. A particular validator uses only the coordinates it needs; for example, an execution check may depend only on (P, o), whereas field novelty necessarily compares (h, o) with time-indexed external prior art (K_{ext}, τ). Validators are coded by separate facets rather than additive layers: target property (e.g., executability, empirical adequacy, novelty, significance), evidence source (execution trace, measurement, replication, literature, review), evaluator substrate (program, statistical model, LLM, human, hybrid), decision role (diagnostic, search control, or acceptance), and external independence (internal, shared, or independent). Multiple values may apply within a facet; values from different facets must not be added or interpreted as maturity levels. An event does not create a route by itself. A separately evidenced edge may route r_j to $D_j \subseteq \{S, G, E, V, M, \pi, A, W_A, \text{external}\}$. Artifact revision is represented specifically as a route to W_A followed by an `artifact_revision` edge that produces a new artifact version A_{v+1}; mutating A_v in place is not an admissible shortcut.

M**: Memory** Structured as (M_e, M_s, M_p). M_e (episodic) stores individual records (h_i, o_i, r_i); M_s (semantic) stores distilled knowledge patterns; M_p (procedural) stores learned protocols. M is the persistent storage and retrieval substrate. It can supply retrieved content to I_t, but it is not identical to the decision-time information state. The structure of M directly determines the quality of G and π.

$\pi : I_t \times M \times \mathcal{C} \times B \times R_{\text{pre}} \to a$**: Search Policy** The decision rule that selects an action $a = (h, d, P, f)$, comprising a candidate $h \in \mathcal{C}$, an experimental design d, an executable protocol P, and a fidelity f. I_t contains only information that the bounded source reports as available at decision time; Servo neither requires a particular state-update map nor reconstructs an unreported latent state. B is the remaining resource budget and R_{pre} contains validation signals routed before action selection. A myopic BED policy holds the candidate and feasible-action construction fixed and selects a (or its design component d) to maximize conditional EIG; this is a one-step action rule, not a globally optimal sequential policy. In practice, systems use reactive, greedy, active-learning, finite-state, recurrent, or tree-search policies. Thus G generates candidates while π chooses what to do with them; an explicitly reported Bayesian or POMDP belief is the special case $I_t = b_t$ [Kaelbling et al., 1998].

Operational event–artifact graph. For a bounded case c, Servo records endpoints, versioned artifacts, evaluation events, and directed edges,

$$\mathcal{G}_c = (\mathcal{N}_c, \mathcal{A}_c, \mathcal{E}_c, \mathcal{R}_c), \tag{3}$$

where $\mathcal{N}_c$ are typed instantiated component or boundary endpoints, $\mathcal{A}_c$ are artifacts with documented production or availability anchors and versions, $\mathcal{E}_c$ are event occurrences, and $\mathcal{R}_c$ are observation, epistemic-update, feedback-control, artifact-revision, or external-only edges. Every member of $\mathcal{E}_c$ has its own `event_id`; a later occurrence requires a distinct event record rather than a time suffix attached to an earlier record. Artifact versions are ordered only within an explicit, case-local `artifact_lineage_id`, so parallel lineages may independently use the same version number. Each event class has an allowed actor component, and each edge type has allowed source–destination component pairs; the package rejects graphs that violate these endpoint-level admissibility constraints. These constraints do not type edge payloads or assert direct composition of the component function signatures. The `event_class` determines actor admissibility under η, whereas `event_kind` records the relation-relevant semantic role of an occurrence. Thus `epistemic_action` is an event kind, not an additional event class; every such occurrence still carries an admissible class such as generation, runtime validation, or state update. Human, system, mixed, or unreported mediation is a coarse actor facet rather than a distinct edge topology; one mediator endpoint does not enumerate every participant in a mixed route. Events and reliability evaluations are different records: an evaluation study is not a runtime event and cannot serve as a feedback edge. A denotes artifact state consumed or produced by events, while the optional W_A denotes a reported authored-artifact synthesis or revision interface. Both lie at the six-component boundary; neither silently expands the core tuple. A terminal event has no outgoing internal edge. Pre-action filtering, a fixed schedule, append-only memory, formatting or copying, retry without epistemic update, and terminal assessment alone are prohibited shortcuts rather than evidence of a feedback relation. The current graph does not instantiate O_{env}, simulator response, or measurement apparatus as endpoints. Its `observation` edge $E \to V$ does not denote causal observation generation: it records that evidence associated with a documented execution became available to a validator. Here E anchors the execution occurrence, not the physical, environmental, simulator, or measurement process that caused the value. Likewise, `producer_event_id` for a result or measurement artifact is a documentary availability anchor within the bounded trace, not necessarily its ultimate causal producer.

The admissibility maps from graph records to tuple or boundary components are explicit. For event class k and edge type r, let $\eta(k)$ be the allowed actor-component set and $\tau(r)$ the allowed ordered component pair set:

$$\begin{aligned}
\eta(\text{runtime validation}) &= \{V\}, & \eta(\text{artifact assessment}) &= \{V\},\\
\eta(\text{generation}) &= \{G\}, & \eta(\text{execution}) &= \{E\},\\
\eta(\text{artifact production}) &= \{W_A\}, & \eta(\text{state update}) &= \{M\},\\
\eta(\text{human feedback}) &= \{\text{external}\}, & \eta(\text{external assessment}) &= \{\text{external}\},
\end{aligned} \tag{4}$$

and

$$
\begin{aligned}
\tau(\text{observation-delivery projection}) &= \{(E, V)\}, \\
\tau(\text{epistemic update}) &= \{(V, M)\}, \\
\tau(\text{artifact revision}) &= \{(V, W_A), (\pi, W_A)\}, \\
\tau(\text{external only}) &= \{(\text{external}, \text{external})\}, \\
\tau(\text{feedback control}) &= \{(V, G), (V, \pi), (M, G), (M, \pi), \\
&\quad (\pi, G), (\pi, E), (G, E), (W_A, E), \\
&\quad (\text{external}, W_A)\}.
\end{aligned} \tag{5}
$$

For every event, the actor component must belong to η of its event class; for every edge, the ordered endpoint-component pair must belong to τ of its edge type. These edges encode source-reported dependency or control routing, not a typed call graph. In particular, $V \to G$ or $\pi \to G$ means that reported information conditions a generation occurrence, while $G \to E$ records delivery of a generated executable candidate or plan after any source-reported selection context; it does not assign action-selection authority to G. The map therefore checks endpoint-level integrity only and does not prove payload compatibility with the abstract function signatures. Unreported mappings remain evidentially unknown.

Typed relations and per-relation evidence. Servo does not assign one yes/partial/full loop label. For each case it records the typed functional relations that the bounded source reports, each as a tuple (*source role*, *relation type*, *target role*, *temporal scope*) over the roles and relation types of the event–artifact graph, and it keeps an evaluation of an observation distinct from the generation of that observation. Every reported relation carries a derived decision claim recorded on separate axes: a *source support* status (supported, unresolved, contradicted, or not applicable), an *occurrence resolution* (resolved, unresolved, or not applicable), and an *evidence mode* naming the kind of proposition the source supports. Source silence lowers occurrence resolution to unresolved rather than to a positive or a not-applicable value, and a relation that crosses an unreported executor–environment measurement boundary is lowered in the same way. Table 5 reports, per case, where observations originate, what evaluation acts on, and whether the source separates an observation from its subsequent evaluation; Table 7 (Appendix C) reports each functional relation with its source support, occurrence resolution, and evidence mode. Evidence modes name the kind of proposition a source supports and are not an ordinal scale of system quality.

Policy decomposition. Applying the POMDP- and BED-derived distinctions to the search policy π, Servo decomposes each case's experiment-selection policy along the selection signal the source names, the information on which follow-up selection depends, the objective that selection serves, and the separate rules by which candidates are selected and executed (Table 4). The objective is kept apart from both candidate rules, so no objective is folded into a rule: the Robot Scientist (C05) carries uncertainty-reduction and hypothesis-discrimination objectives yet an exhaustive selection rule and an all-selected execution rule—it tested every generated hypothesis rather than performing an expected-discrimination down-selection—whereas the score-cluster cases (C02–C04, C06) pursue performance improvement with ranked or thresholded selection, and Coscientist (C01) reacts to execution failure and reaction yield without a score dependence. Read at the level of individual occurrences, the feedback relations established at occurrence-mode evidence are few: only two carry occurrence-mode evidence—Coscientist's evaluation-to-artifact revision and the Robot Scientist's evaluation-to-inquiry-state update—while the one evaluation-to-execution feedback relation (AI Scientist 2024) is supported only at aggregate evidence, so no case establishes an evaluation-to-execution relation at occurrence-mode evidence. None of the bounded sources reports an explicit posterior-based or expected-information-gain criterion for experiment selection; this is a corpus-level reporting observation, not evidence of a field-wide absence.

Human involvement is recorded as a categorical authority vector rather than the former scalar $H \in [0, 1]$: $H = (H_S, H_G, H_E, H_V, H_M, H_\pi)$ gives one coarse component-level label—human control, human participation, system control, shared control, or not reported. It does not distinguish initiation, advice, approval, veto, direct execution, and final authority within the same component. H_S separately identifies coarse authority over S_0 and later space expansion from candidate generation by G. The vector is descriptive, not numeric or ordered [Parasuraman et al., 2000].

Faceted validator description. The former labels V_{syntax}, V_{semantic}, $V_{\text{empirical}}$, V_{human}, and V_{formal} mixed target properties, evidence sources, evaluator substrates, and decision roles; they were neither

ID	Lineage	System/version	Configuration	Task regime
C01	coscientist	Coscientist (2023 paper)	Reported demonstration	Six chemistry tasks
C02	ai_scientist	AI Scientist 2024 (2024 paper)	Reported default	At most five computational experiments
C03	ai_scientist	AI Scientist 2026 (2026 Nature)	Template-free tree	Computational tree search
C04	agent_laboratory	Agent Laboratory (2025 paper)	Autonomous mode	Human-provided ML idea
C05	robot_scientist	Robot Scientist (2010 review of Adam)	Adam yeast	Yeast functional genomics
C06	novelseek	InternAgent (formerly NovelSeek) (2025 InternAgent)	Reported multi-agent	Twelve computational tasks

Table 3: Six versioned cases from five lineages. IDs, versions, configurations, and task regimes are generated from the canonical Servo records.

mutually exclusive layers nor a reliability order. Gate reliability evidence is recorded separately from runtime events and is called probabilistic calibration only when confidence–outcome correspondence is measured. Balanced accuracy, false-positive rate, mean score bias, and coder agreement are not calibration. The superseded binary fields remain reconstruction data only and are not inputs to the current schema.

The validator as a reward channel. Treating V as distinct from the observation likelihood, Appendix A records three analytical separations. Posterior consistency belongs to the observation model and data-generating regime; a validator changes the inferential target only when it is coupled into a properly normalized update likelihood (or indirectly changes future designs); and a novelty component invisible to every available channel is non-identifiable. These observations diagnose where validator design can matter. They do not establish that any scalar validator measure is universally necessary or sufficient for trustworthy autonomous discovery.

5 Analysis of Core AI Scientist Systems

The frozen application contains six versioned cases from five lineages: Coscientist; AI Scientist 2024 and AI Scientist 2026 as two versions in one lineage; Agent Laboratory; Robot Scientist; and InternAgent (formerly NovelSeek). Each case is bounded by a reported configuration and task regime. The records are selected source-traceable examples, not "core" representatives of a population. They describe S, G, E, V, M, π, the reporting status of the conceptual O_{env} boundary, the recorded boundary objects A, W_A, human authority, endpoints, events, artifacts, routed edges, reliability evaluations, and the typed functional relations recorded under the Servo event–artifact contract.

Every relation is keyed by `case_id` and its typed `(source_role, relation_type, target_role, temporal_scope)` tuple in the canonical Servo records. Each relation resolves the event, edge, endpoint, artifact, evidence, source-location, and source-hash records in the release package that carry its per-relation support status and occurrence resolution.

The event graph preserves distinctions that a bare loop label would erase. In the AI Scientist lineage, pre-action literature filtering, runtime experiment or tree evaluation, manuscript assessment, and external review are separate events with different artifacts and destinations [Lu et al., 2024, 2026]. Coscientist reports execution and artifact-revision relations, while Agent Laboratory reports execution and artifact-evaluation relations; each relation is recorded only where the source supports it and its occurrence resolution is stated within the case [Boiko et al., 2023, Schmidgall et al., 2025]. Robot Scientist reports a physical assay and model-update cycle within its bounded yeast-functional-genomics regime [King et al., 2004, Sparkes et al., 2010]; InternAgent reports multi-agent assessment, debugging, and evolution while Servo deliberately compresses agent-to-agent organization [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025]. Human scope, manual execution, terminal authority, and external assessment remain separately recorded rather than being converted into a scalar autonomy value.

The normative data model consists of the case, endpoint, artifact, event, edge, reliability-evaluation, functional-relation, derived-decision-claim, domain-anchor, and selection-ledger records named in

Case	Selection signal	Follow-up (control) dependence	Selection-purpose facets	Candidate-selection rule	Candidate-execution rule
C01	software/execution error reaction yield prior-round observations normalized advantage value	`failure` `observation_result`	`local_repair` `performance_improvement`	`sequential_choice`	`one_at_a_time` `until_success`

C02	execution/compilation error
	self-assessed idea scores (interestingness/novelty/feasibility)
	automated review scores
	experiment results
	literature-similarity to existing work
		`failure`
		`score`
		`observation_result`
			`local_repair`
			`performance_improvement`
			`diversity_directed_selection`
				`threshold`	`one_at_a_time`
					`until_budget`

C03	execution error / buggy node
	VLM-flagged plot issues
	LLM-evaluator performance metrics (training dynamics, plots)
	best-performing checkpoint score
	training-curve convergence
		`failure`
		`score`
		`observation_result`
			`local_repair`
			`performance_improvement`
				`top_k_ranked`
				`sampled_subset`
					`batch`
					`until_budget`

C04	subtask completion (unconditional autonomous progression)
	compile/runtime error
	mle-solver alignment score
	professor-agent reward-model score (0-1)
	experiment results
		`fixed_or_predefined`
		`failure`
		`score`
		`observation_result`
			`local_repair`
			`performance_improvement`
				`top_k_ranked`
				`sampled_subset`
					`batch`
					`until_success`
					`until_budget`

C05	fixed four-step orphan-enzyme method
	sequence similarity (PSI-BLAST/FASTA)
	confirm/refute statistical outcome
	model conflicts flagged for update
		`fixed_or_predefined`
		`score`
		`observation_result`
			`uncertainty_reduction`
			`hypothesis_model_discrimination`
				`exhaustive`	`all_selected`

C06	runtime exceptions
	task complexity (coder-module routing)
	assessment-agent idea score (0-10, five dimensions, weighted sum)
	structured critiques (automated + human feedback)
	empirical performance results
		`failure`
		`score`
		`observation_result`
			`local_repair`
			`performance_improvement`
				`top_k_ranked`	`until_success`
					`until_budget`

Legend. Schema-v3 BED-lens decomposition of each case's experiment-selection policy (charter B.4 / contract section B); an analytic decomposition, not a compliance label. Each cell lists the enum values the bounded source reports for that axis; a multi-valued axis is a set, not an ordering. The selection-purpose facets (the objective) are kept separate from the candidate-selection and candidate-execution rules, so no objective is folded into a rule. Two policy axes are recorded in the source data but are not columns here: *generation scope* (`fixed_space` / `search_space_expansion` / `candidate_diversification`) is a G/S-level provenance axis, and *formal epistemic-utility evidence* is `not_reported` for all six cases. *Formal epistemic-utility evidence.* None of the bounded sources reports an explicit posterior or expected-information-gain criterion; this is a corpus-level reporting observation, not evidence of field-wide absence.

Table 4: Policy decomposition (Table A): a schema-v3 BED-lens reading of each case's experiment-selection policy (columns and axis domains detailed in the legend). The selection-purpose facet (the objective) is kept apart from both candidate rules, so no objective is folded into a rule: C05 (Robot Scientist) carries `uncertainty_reduction` and `hypothesis_model_discrimination` objectives yet `exhaustive` selection and `all_selected` execution rules (it tested every generated hypothesis, not an expected-discrimination down-selection), whereas the score-cluster cases (C02–C04, C06) are `performance_improvement` with `top_k_ranked`/`threshold`/`sampled_subset` selection, and C01 reacts to failure and observation results without a `score` dependence (`sequential_choice` selection). This is an analytic decomposition, not a compliance label.

Case	Observation source	Evaluation target	O–V boundary
C01	`environment`	`candidate` `observation`	`separated`

C02	`observation`	`artifact`
		`candidate`
		`observation`
			`separated`

C03	`environment`	`artifact`
		`candidate`
		`execution`
		`observation`
			`separated`

C04	`observation`	`artifact`
		`candidate`
		`observation`
			`separated`

C05	`environment`	`candidate` `observation`	`separated`

C06	`environment`	`artifact`
		`candidate`
		`observation`
			`separated`

Legend. Case-level observation–evaluation boundary (contract section C). *Observation source* = the role(s) the bounded source names as producing an observation (· →`produces`→`observation`); `observation` as a source denotes a within-measurement transform whose upstream environment/execution origin the source leaves unreported (the E–O_env boundary is `boundary_unreported`). *Evaluation target* = the role(s) an `evaluation`→`evaluates`→ · relation acts on. *O–V boundary* ∈ {`separated`, `boundary_unreported`, `merged`, `n.a.`} reads the O_env–V split carried by the `evaluation`→`observation` relations (rubric v5-rubric-3): `separated` = the source reports that split (a distinct evaluation of a distinct observation) in at least one relation; `boundary_unreported` = such relations exist but none reports the split; `merged` = observation and evaluation recombined into one step; `n.a.` = no evaluation-of-observation relation. In this corpus `merged` and `n.a.` do not occur.

Table 5: Observation–evaluation boundary (Table B1), read at case level. For each case the table reports where observations originate, what evaluation acts on, and whether the source separates an observation from its subsequent evaluation. All six bounded sources report that split (`separated`) in at least one relation, populated from the source rather than assumed.

`servo_v5_schema.yaml`. The frozen 42-session, three-vendor multicoder audit used a superseded record-level rubric and did not code the current event, artifact, edge, or relation units; it is retained only as a development audit.

5.1 Contrastive framework readings

The crosswalk in Table 2 is made concrete by three contrasts. First, a single loop label for AI Scientist (Nature 2026) hides the difference between stage and tree evaluators, a manuscript reviewer, and workshop review after manual submission selection. Second, Agent Laboratory's internal execution path is distinct from its paper-quality assessment and retained human authority. Third, InternAgent's native multi-agent architecture represents agent roles and communication more directly than Servo's system-level case record [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025]. These are differences in foregrounded units, not a performance or superiority comparison.

The Robot Scientist is a negative case against an automatic claim of added diagnostic value: its source already exposes the architecture-level hypothesis, assay, update, and selection sequence [King et al., 2004, Sparkes et al., 2010]. It does not, however, identify a distinct post-update execution and evidence occurrence, so the stricter occurrence-level adaptation and discovery relations remain unresolved. Servo supplies a consistently validated record format and a documented mapping target, but does not thereby discover a hidden mechanism. The examples show only what the schema foregrounds.

5.2 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem—what the *output* of the loop should be and how to verify it. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-part ontology (`/logic`, `/src`, `/trace`, `/evidence`). In the final Servo facets, its Seal targets schema

ID	System/domain	Analysis target	Evaluator/evidence	Decision role	Fidelity boundary
DA01	FunSearch / Formal mathematics	Program quality and task performance	Executable evaluator	Ranking and search control	Computational oracle
DA02	AI Feynman / Physics	Symbolic fit to data	Symbolic-regression algorithm	Model search and fit	Computational data fit
DA03	ChemCrow / Chemistry	Task performance and synthesis	Tools, execution, and assessment	Planning, execution, assessment	Computational and physical
DA04	GNoME / Materials science	Phase stability and energy	Graph networks plus DFT	Filtering and active-learning update	Surrogate; separate experiment
DA05	BioPlanner / Biology	Protocol-planning pseudocode reconstruction	Generator; pseudocode evaluator; one-off lab check	Diagnostic benchmark; single validation	Protocol; single lab run
DA06	Manning automated social science / Social science	Causal hypotheses and simulated effects	Causal model and LLM agents	Design and model update	Simulated subjects
DA07	AutoML-Zero / Computer science	Evolved-algorithm benchmark performance	Executable benchmark	Selection and evolutionary update	Computational benchmark

Table 6: Seven selected, frozen source anchors. The rows demonstrate how the vocabulary can be instantiated at named boundaries; they are not representative domain samples or a common outcome scale.

compliance, argumentative rigor, and reproducibility using deterministic checks, an automated Rigor Auditor (82.6% recall on a seeded rigor-defect benchmark), and sandboxed execution; significance and novelty remain assigned to human judgment [Liu et al., 2026]. Motivating figures: only 45.4% of reproduction requirements are fully specified, and below-reference agent runs account for 90.2% of total dollar cost—exploration that becomes waste only when later agents cannot reuse it [Liu et al., 2026]. The `/trace` part externalizes M_e, with distilled semantic content (M_s) held in `/logic`; preserved traces both accelerated and anchored downstream agents—richer M_e does not monotonically improve π.

6 Domain Analysis

Table 6 lists seven selected, frozen source anchors. Each row names one system and one bounded primary source; rows are not merged into a domain-level case, sampling frame, prevalence claim, or cross-domain outcome comparison.

The anchor rows support only source-local descriptions. They do not establish that a pattern holds across a domain, that one validator type causes an outcome, or that the seven anchors cover the relevant design space. DFT approximates physical energies rather than serving as a physical oracle, and predicted compounds still require synthesis and characterization [Cheetham and Seshadri, 2024].

7 Open Problems

The selected case and anchor sources motivate three prospective design questions. They do not establish prevalence or field-wide absence, and none of the proposed evaluations is executed here.

7.1 The Lack of Validated Automated Novelty Gates

The bounded sources used here do not report an independently validated automated novelty gate. This is a source-limited evidence statement, not a finding that every automated novelty score is unreliable or that no other system supplies such evidence. The cited failures concern adjacent properties: paper-quality bias, hallucinated tables, proof correctness, structural violations, or review overlap. They motivate caution but do not measure novelty discrimination. Novelty, correctness, significance, and aggregate review quality remain separate endpoints.

Prospective validation design requirements: These are necessary design requirements, not a sufficient checklist for declaring a gate validated. A preregistration must define the target population, temporal cutoff, domain strata, novelty prevalence or sampling scheme, primary discrimination endpoint (for example sensitivity/specificity or AUC), minimally relevant performance, and confidence-interval width. A prospective power or precision calculation must then determine benchmark-item and rater counts per domain; “three domains” and “five experts” may serve as diversity floors but never substitute for that calculation. Experts label independently, human–human agreement is reported with uncertainty, and held-out adjudicated field-novelty labels use explicit prior-art search kept separate from significance and correctness. Follow-up duration, censoring, and the rule for later priority or prior-art outcomes must also be preregistered. The automated system is evaluated against held-out adjudicated labels rather than the majority vote from the same raters. High agreement with a small panel or separation of published from withheld findings remains only an initial benchmark result. A recent benchmark takes a step by restricting to findings from 2024 onward to control contamination [Liu et al., 2025], though it evaluates inspiration-based decomposition rather than satisfying this design.

Novelty, significance, and correctness are distinct properties that the bounded sources may score separately and then combine at a decision point (The AI Scientist's paper reviewer scores separate originality, quality, clarity and significance axes, yet collapses them into an aggregate `Overall` score and accept/reject decision [Lu et al., 2024]). Functional separation of generation from reviewing, reflection, or ranking does not imply statistical or organizational independence. The selected sources do not report independent, contamination-controlled evidence that such a channel discriminates field novelty.

7.2 The BED–Practice Gap for π

Equation 1 specifies a one-step, myopic information criterion. The bounded LLM-case sources used here do not state that their policies optimize EIG or a sequential information objective. Modern BED can use variational estimators, amortized policies, and explicit probabilistic surrogates [Foster et al., 2019, 2021]; BED-LLM [Choudhury et al., 2026] provides a separate example for conversational queries. The open question is integration with a reported G–π action interface and scientific evidence path, not an asserted field-wide absence.

Prospective success criterion: A deployed AI Scientist narrows the survey gap if it maintains an observation-updated probabilistic model—exact, variational, ensemble, or other documented surrogate—and uses a validated EIG or sequential information objective to select actions a_t under explicit budgets. Evaluation should compare discovery-relevant utility and information efficiency against matched heuristic policies. Direct Bayesian representation of open-ended natural-language or program hypotheses is a separate, stronger representation problem; it is not required to count a practical BED integration.

Architecturally, a system may maintain an explicit probabilistic surrogate or use variational and amortized EIG estimation [Foster et al., 2019, 2021]. What remains difficult in open-ended $\mathcal{S}$ is defining and updating a representation on which those objectives are meaningful. EIG-based selection presupposes an already represented hypothesis space, whereas how genuinely new hypotheses or representations *arise* is a separate generative problem [Yanai and Lercher, 2019].

7.3 The Multi-Fidelity Integration Gap

Test-time compute has created a structural asymmetry: hypothesis generation is now fast and cheap, while experiment execution E remains physically rate-limited. Principled criteria for selecting among costly information sources already exist in multi-fidelity Bayesian optimization: continuous-fidelity models trade approximation cost against target information [Kandasamy et al., 2017], and cost-sensitive mutual-information acquisition selects among simulations and real evaluations [Song et al., 2019]. The gap identified here is therefore not the absence of mathematical escalation criteria. It is their unvalidated integration into the surveyed open-ended AI Scientist workflows, where fidelity changes may alter protocols, outcome semantics, and physical feasibility.

The selected anchors expose several proxy-to-physical boundaries. GNoME predicts 380,000 stable compounds via DFT [Merchant et al., 2023]; 736 match structures independently present in or added to the ICSD, including 184 added after the project began. These are concurrent records by others rather than syntheses the system triggered. ChemCrow reports wet-lab execution, whereas physical characterization remains protocol-specific [Bran et al., 2024, Gromski et al., 2019]. BioPlanner evaluates LLM-generated protocol pseudocode without thereby establishing a biological-validity feedback witness [O'Donoghue et al., 2023]. These anchors motivate, but do not establish the prevalence of, a multi-fidelity integration problem.

This design problem is distinct from Op1 and Op2: a proposed implementation would need to make evidence comparable across fidelities and expose fidelity and cost to π. Existing MFBO supplies acquisition criteria under stated surrogate assumptions; whether those assumptions and protocol transitions hold in an AI Scientist workflow is an empirical question.

Autonomous laboratories such as A-Lab [Szymanski et al., 2023], ORGANA [Darvish et al., 2024], and AutoLabs [Panapitiya et al., 2025] show that execution automation and fidelity selection are separate engineering questions. They are cited as adjacent examples, not as an exhaustive audit of multi-fidelity policy evaluation.

Prospective success criterion: A system closes the integration gap if it implements a documented multi-fidelity acquisition rule, including cost and cross-fidelity uncertainty, and prospectively outperforms matched fixed-fidelity or fixed-escalation baselines on experimentally validated discoveries per unit physical cost. This tests adoption and calibration in an AI Scientist workflow rather than claiming to invent the underlying criterion.

8 Discussion

Validator design as a diagnostic vocabulary. Unreliable validation can distort what a system records as progress—an ecosystem-scale symptom is the documented surge in fabricated citations [Zhao et al., 2026], and strong task performance need not imply an internalized world model [Vafa et al., 2025]. The earlier gate-reliability/"trustworthy closure" association is withdrawn as conceptually circular, not merely left untested. The framework instead asks property-specific questions: which channel fires when, where its output is routed, what property it assesses, what bias or calibration evidence exists, at what experimental fidelity, and with what external independence? Any future outcome analysis must use gate-independent external endpoints Y_q.

Human oversight. The categorical authority vector localizes coarse human or system control by component; it does not resolve whether participation means initiation, advice, approval, veto, direct execution, or final authority. It is not a quantity to minimize. Human authority in novelty/significance assessment and policy is appropriate for the surveyed workflows because their automated proxies do not yet provide independently validated judgments of those targets. This is a current-workflow finding, not a claim that those judgments are non-automatizable in principle [Parasuraman et al., 2000].

Towards collective AI co-scientists. Some analyzed systems, including InternAgent, are explicitly multi-agent [InternAgent Team, Shanghai Artificial Intelligence Laboratory, 2025]; this paper nevertheless collapses each implementation into one system-level (S, G, E, V, M, π) tuple and does not model agent-level roles, communication, disagreement, or authority. The relevant extension is therefore from that system-level abstraction to an agent graph, shared-memory process, collective validator, or Dec-POMDP-like model, rather than the introduction of multi-agent systems as such. Collective Predictive Coding [Taniguchi et al., 2025] and ARA [Liu et al., 2026] offer possible starting points for shared epistemic state. Such an extension must preserve diversity: AI reviewers already overlap more than independent humans [Kim et al., 2026], risking the transient diversity that benefits epistemic communities [Zollman, 2010].

Limitations. Servo is non-exhaustive and abstracts away agent-level and social organization [Taniguchi et al., 2025]. It is not a complete ontology, causal theory, population taxonomy, or performance ranking. The current graph does not instantiate O_{env}, simulator response, or measurement apparatus as endpoints or activities. It therefore cannot distinguish execution failure from environment or measurement failure, compare observation laws attached to the same executor, or provide lossless causal provenance for measurement artifacts. Its $E \to V$ observation edge and result-artifact producer fields preserve only a source-bounded execution anchor and subsequent evidence availability. The six cases informed framework development, while the seven anchors are partial records rather than complete held-out graphs. No untouched case has yet been instantiated end-to-end under a prospectively frozen contract. The present evidence therefore supports internal consistency and a loss-aware standards mapping, not demonstrated out-of-sample portability; a future holdout that triggers a schema change must be reclassified as formative. A per-relation evidence claim concerns only its bounded case and relation and does not imply scientific outcome quality. Servo supplies neither a causal model nor a generative law for missing routes, costs, fidelity transitions, or artifacts. The frozen 42-session, three-vendor multicoder audit predates the current event–artifact contract and is retained only as a historical development audit; its agreement coefficients do not establish reproducibility, source entailment, construct validity, human reliability, real-world utility, or comparative superiority.

9 Conclusion

We introduced Servo (S, G, E, V, M, π) as a non-exhaustive, source-traceable event-and-evidence schema. Its operational graph is a documentary projection rather than a complete causal provenance model: the tuple distinguishes the executor from O_{env}, while the released graph records only source-supported execution anchors and evidence availability at evaluation events. Applying Servo's POMDP- and BED-derived distinctions decomposes six cases drawn from a literature that commonly uses terms such as iterative or closed-loop, by separating observation from evaluation and by distinguishing the information on which follow-up selection depends, the objective that selection serves, and the rule determining which candidates are executed: some policies react to yield or evaluation scores, while one case (C05, the Robot Scientist) reports testing all generated hypotheses in a discrimination-oriented procedure. As a closing corpus-level observation, none of the bounded sources reports an explicit posterior-based or expected-information-gain criterion for experiment selection—a limitation of the bounded public sources analyzed here, not an asserted field-wide absence. These formative cases do not establish out-of-sample portability, prevalence, causality, comparative superiority, or scientific-outcome effects.

References

R. H. Berk. Limiting behavior of posterior distributions when the model is incorrect. *The Annals of Mathematical Statistics*, 37(1):51–58, 1966.

T. Blau, E. V. Bonilla, I. Chades, and A. Dezfouli. Optimizing sequential experimental design with deep reinforcement learning. In *International Conference on Machine Learning*, 2022. arXiv:2202.00821.

D. A. Boiko, R. MacKnight, B. Kline, and G. Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.

A. M. Bran, S. Cox, O. Schilter, C. Baldassari, A. D. White, and P. Schwaller. ChemCrow: Augmenting large-language models with chemistry tools. *Nature Machine Intelligence*, 2024. arXiv:2304.05376.

K. Chaloner and I. Verdinelli. Bayesian experimental design: A review. *Statistical Science*, 10(3): 273–304, 1995.

A. K. Cheetham and R. Seshadri. Artificial intelligence driving materials discovery? Perspective on the article: Scaling deep learning for materials discovery. *Chemistry of Materials*, 36, 2024. doi: 10.1021/acs.chemmater.4c00643. Available: `https://escholarship.org/content/qt9qx9t3kz/qt9qx9t3kz.pdf`.

D. Choudhury, S. Williamson, A. Goliński, N. Miao, F. B. Smith, M. Kirchhof, Y. Zhang, and T. Rainforth. BED-LLM: Intelligent information gathering with LLMs and Bayesian experimental design. In *International Conference on Learning Representations*, 2026. arXiv:2508.21184.

K. Darvish, M. Skreta, Y. Zhao, N. Yoshikawa, S. Som, M. Bogdanovic, Y. Cao, H. Hao, H. Xu, A. Aspuru-Guzik, A. Garg, and F. Shkurti. ORGANA: A robotic assistant for automated chemistry experimentation and characterization, 2024. arXiv:2401.06949; also in Matter (Cell Press) 2025.

T. Feng, T. H. Trinh, G. Bingham, D. Hwang, Y. Chervonyi, et al. Towards autonomous mathematics research. *arXiv preprint arXiv:2602.10177*, 2026. arXiv:2602.10177.

A. Foster, M. Jankowiak, E. Bingham, P. Horsfall, Y. W. Teh, T. Rainforth, and N. Goodman. Variational Bayesian optimal experimental design. In *Advances in Neural Information Processing Systems*, 2019. arXiv:1903.05480.

A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth. Deep adaptive design: Amortizing sequential Bayesian experimental design. In *International Conference on Machine Learning*, 2021. arXiv:2103.02438.

S. Ghosal and A. van der Vaart. *Fundamentals of Nonparametric Bayesian Inference*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2017.

J. Gottweis, W.-H. Weng, A. Daryin, T. Tu, P. Sirkovic, A. Myaskovsky, G. Glowatyj, F. Weissenberger, A. Orlandi, D. Popovici, et al. Accelerating scientific discovery with Co-Scientist. *Nature*, 2026. doi: 10.1038/s41586-026-10644-y. arXiv:2502.18864.

P. S. Gromski, A. B. Henson, J. M. Granda, and L. Cronin. How to explore chemical space using algorithms and automation. *Nature Reviews Chemistry*, 3:119–128, 2019. doi: 10.1038/s41570-018-0066-y.

InternAgent Team, Shanghai Artificial Intelligence Laboratory. InternAgent: When agent becomes the scientist – building a closed-loop system from hypothesis to verification. *arXiv preprint arXiv:2505.16938*, 2025. arXiv:2505.16938v3. Circulated as NovelSeek in earlier versions; the key is retained for continuity with prior citations of this work.

L. P. Kaelbling, M. L. Littman, and A. R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1–2):99–134, 1998.

K. Kandasamy, G. Dasarathy, J. Schneider, and B. Póczos. Multi-fidelity Bayesian optimisation with continuous approximations. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1799–1808, 2017.

S. Kim, D. Yoon, K. Gashteovski, J. Suk, J. Baek, P. Aggarwal, I. Wu, et al. On the limits and opportunities of AI reviewers: Reviewing the reviews of Nature-family papers with 45 expert scientists. *arXiv preprint arXiv:2605.20668*, 2026.

R. D. King, K. E. Whelan, F. M. Jones, P. G. K. Reiser, C. H. Bryant, S. H. Muggleton, D. B. Kell, and S. G. Oliver. Functional genomic hypothesis generation and experimentation by a robot scientist. *Nature*, 427(6971):247–252, 2004. doi: 10.1038/nature02236.

J. Liu et al. The last human-written paper: Agent-Native research artifacts. *arXiv preprint arXiv:2604.24658*, 2026. arXiv:2604.24658.

Y. Liu, Z. Yang, T. Xie, J. Ni, B. Gao, Y. Li, S. Tang, W. Ouyang, E. Cambria, and D. Zhou. ResearchBench: Benchmarking LLMs in scientific discovery via inspiration-based task decomposition, 2025. arXiv:2503.21248; Findings of ACL 2026.

C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha. The AI Scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.

C. Lu, C. Lu, R. T. Lange, Y. Yamada, S. Hu, J. Foerster, D. Ha, and J. Clune. Towards end-to-end automation of AI research. *Nature*, 651:914–919, 2026. doi: 10.1038/s41586-026-10265-5.

B. S. Manning, K. Zhu, and J. J. Horton. Automated social science: Language models as scientist and subjects. Working Paper 32381, National Bureau of Economic Research, 2024. arXiv:2404.11794.

A. Merchant, S. Batzner, S. S. Schoenholz, M. Aykol, G. Cheon, and E. D. Cubuk. Scaling deep learning for materials discovery. *Nature*, 624:80–85, 2023.

L. Moreau and P. Missier. PROV-DM: The PROV data model. W3c recommendation, World Wide Web Consortium, 2013. URL `https://www.w3.org/TR/prov-dm/`.

O. O'Donoghue, A. Shtedritski, J. Ginger, R. Abboud, A. E. Ghareeb, J. Booth, and S. G. Rodriques. BioPlanner: Automatic evaluation of LLMs on protocol planning in biology. *arXiv preprint arXiv:2310.10632*, 2023. EMNLP 2023.

G. Panapitiya, E. Saldanha, H. Job, and O. Hess. AutoLabs: Cognitive multi-agent systems with self-correction for autonomous chemical experimentation, 2025. arXiv:2509.25651.

R. Parasuraman, T. B. Sheridan, and C. D. Wickens. A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics—Part A: Systems and Humans*, 30(3):286–297, 2000. doi: 10.1109/3468.844354.

T. Rainforth, A. Foster, D. R. Ivanova, and F. B. Smith. Modern Bayesian experimental design. *Statistical Science*, 2024. arXiv:2302.14545.

E. Real, C. Liang, D. So, and Q. Le. AutoML-Zero: Evolving machine learning algorithms from scratch. In *International Conference on Machine Learning*, 2020. arXiv:2003.03384.

B. Romera-Paredes, M. Barekatain, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. Ellenberg, P. Wang, O. Fawzi, et al. Mathematical discoveries from program search with large language models. *Nature*, 625:468–475, 2024.

S. Schmidgall, Y. Su, Z. Wang, X. Sun, J. Wu, X. Yu, J. Liu, M. Moor, Z. Liu, and E. Barsoum. Agent Laboratory: Using LLM agents as research assistants. *arXiv preprint arXiv:2501.04227*, 2025.

J. Song, Y. Chen, and Y. Yue. A general framework for multi-fidelity Bayesian optimization with Gaussian processes. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 3158–3167, 2019.

A. Sparkes, W. Aubrey, E. Byrne, A. Clare, M. N. Khan, M. Liakata, M. Markham, J. Rowland, L. N. Soldatova, K. E. Whelan, M. Young, and R. D. King. Towards robot scientists for autonomous scientific discovery. *Automated Experimentation*, 2(1):1, 2010. doi: 10.1186/1759-4499-2-1.

N. J. Szymanski, B. Rendy, Y. Fei, R. E. Kumar, T. He, D. Milsted, M. J. McDermott, M. Gallant, E. D. Cubuk, A. Merchant, H. Kim, A. Jain, C. J. Bartel, K. Persson, Y. Zeng, and G. Ceder. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624:86–91, 2023. doi: 10.1038/s41586-023-06734-w.

T. Taniguchi, S. Takagi, J. Otsuka, Y. Hayashi, and H. T. Hamada. Collective predictive coding as model of science: Formalizing scientific activities towards generative science. *Royal Society Open Science*, 12(6):241678, 2025. doi: 10.1098/rsos.241678. arXiv:2409.00102.

G. Tie, P. Zhou, and L. Sun. A survey of AI scientists: Surveying the automatic scientists and research. *arXiv preprint arXiv:2510.23045*, 2025.

S.-M. Udrescu and M. Tegmark. AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631, 2020.

K. Vafa, P. G. Chang, A. Rambachan, and S. Mullainathan. What has a foundation model found? Using inductive bias to probe for world models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2507.06952.

J. Wei, Y. Yang, X. Zhang, Y. Chen, X. Zhuang, Z. Gao, D. Zhou, G. Wang, et al. From AI for science to agentic science: A survey on autonomous scientific discovery. *arXiv preprint arXiv:2508.14111*, 2025. arXiv:2508.14111v2.

Workflow Run RO-Crate Community. Workflow run RO-Crate profile. `https://www.researchobject.org/workflow-run-crate/profiles/workflow_run_crate/`, 2024. Specification, version 0.5, published 19 June 2024 (Zenodo record 12159311).

I. Yanai and M. Lercher. Night science. *Genome Biology*, 20(1):179, 2019. doi: 10.1186/s13059-019-1800-6.

Z. Zhao, Y. Wang, T. Stuart, M. De Vaan, P. Ginsparg, and Y. Yin. LLM hallucinations in the wild: Large-scale evidence from non-existent citations. *arXiv preprint arXiv:2605.07723*, 2026.

K. J. S. Zollman. The epistemic benefit of transient diversity. *Erkenntnis*, 72(1):17–35, 2010.

Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research. The checklist has two parts: a **Research Stage Assessment** (items 1–4) rating AI involvement and iteration effort, and an **AI System Documentation** section (items 5–6).

Research Stage Assessment

1. **Hypothesis development.**

   Answer: **[B]**

   Iteration Effort: **[Medium]**

   Explanation: The core Servo framework tuple (S, G, E, V, M, π), the original V-completeness thesis (now retained only as correction history), the human-authority representation, and the three open problems were conceived by the human author. A frontier LLM agent elaborated framework components, identified literature-level instantiations, and proposed theoretical connections. The human author directed all conceptual decisions and made several substantive corrections to the AI's elaborations.

2. **Experimental design and implementation.**

   Answer: **[B]**

   Iteration Effort: **[Medium]**

   Explanation: This submission contains no computational experiments. The methodology for cross-system analysis (system selection criteria, component extraction protocol, domain comparison structure) was designed by the human author. A frontier LLM agent executed literature retrieval, extracted framework components from primary sources, and performed citation-level fact verification across four verification sweeps.

3. **Analysis of data and interpretation of results.**

   Answer: **[C]**

   Iteration Effort: **[Medium]**

   Explanation: The LLM agent extracted Servo components from primary sources and performed citation-level fact verification. The human author reviewed and corrected the annotations. The six case records and seven source anchors are an author-directed synthesis with source provenance; there is no independent human gold standard, and historical agreement statistics are not presented as evidence of reliability or construct validity for the current schema.

4. **Writing.**

   Answer: **[C]**

   Iteration Effort: **[High]**

   Explanation: The LLM agent drafted the majority of the manuscript text through multiple revision cycles; the human author provided key conceptual passages (framework definitions in §4, core open problems in §7), directed structural decisions, reviewed all drafts, and made final editorial judgments.

AI System Documentation

5. **AI systems used.**

   Description: A frontier large language model with code execution and long-document processing capabilities was used as the primary research and writing assistant. The system operated as an interactive CLI agent with file-system access, able to read primary sources, execute and debug Python simulations, and iteratively revise LaTeX manuscripts.

6. **Observed AI Limitations.**

   Description: (1) Systematic citation fabrication: the agent generated plausible but non-existent quotes and statistics, requiring four dedicated citation-verification sweeps using parallel sub-agents. (2) Analytical overconfidence: the agent initially overclaimed empirical significance for mechanically-entailed theoretical consequences; multiple adversarial review rounds were needed to correct framing and remove overclaims. (3) Context drift:

over long sessions, the agent occasionally regressed previously corrected errors, requiring explicit re-verification of integrity items before each commit.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the non-exhaustive Servo schema contribution, its six bounded cases and seven frozen anchors, and the prospective open questions. Appendix A supplies bounded analytical observations rather than a new theorem, while the case and anchor records are explicitly non-representative source annotations.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Limitations are discussed in §8 (framework abstracts domain cost structures, social layer, non-stationarity) and throughout the domain analysis (§6).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [NA]

   Justification: The paper does not claim a new theorem or proof. Appendix A states three bounded analytical observations, identifies the extra conditions needed to invoke standard results on posterior consistency and model misspecification, and expressly leaves adaptive open-ended regimes outside their scope. The selected source anchors illustrate vocabulary use rather than serving as a proof.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: This paper contains no computational experiment, but its main case annotations and per-relation evidence claims depend on the Servo v5 event–artifact package—the schema, the canonical source-proposition, alignment, and derived-decision-claim records, the policy decomposition, the generated tables, the evidence ledger, and the public regression tests—provided as the supplementary material accompanying this revised manuscript. An earlier, checksum-bound release archives the superseded version at `https://github.com/ysys143/servo-caisc2026/releases/tag/servo-corrected-v4.1.0`; it predates the v5 per-relation reconstruction and is retained only as the prior record, so its closure-predicate artifacts are not the basis of the analysis reported here.

5. **Open access to data and code**

   Question: Does the paper provide open access to data and code?

   Answer: [Yes]

   Justification: The supplementary material binds the current case, endpoint, artifact, event, edge, reliability, functional-relation, derived-decision-claim, domain-anchor, selection-ledger, contract, code, generated tables, audit record, and corrected manuscript files.

6. **Experimental setting/details**

   Question: Does the paper specify all details necessary to understand the results?

   Answer: [Yes]

   Justification: The cross-domain analysis methodology is described in §5 and §6. No computational experiments were performed in this submission.

7. **Experiment statistical significance**

Question: Does the paper report error bars or statistical significance information?

Answer: [NA]

Justification: This paper contains no computational experiments and reports no statistical results requiring error bars or significance tests.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on compute resources?

   Answer: [NA]

   Justification: This submission contains no computational experiments. The meta-analysis requires only standard document processing and no specialized compute.

9. **Research Ethics**

   Question: Does the research conform with the NeurIPS Code of Ethics, except where CAISc's AI authorship policies apply?

   Answer: [Yes]

   Justification: This is a meta-analysis of published systems with no human subjects, no proprietary data, and no safety risks. AI authorship is documented per CAISc policy.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive and negative societal impacts?

    Answer: [Yes]

    Justification: §8 addresses both the potential for accelerating scientific discovery and the risk of premature convergence under widespread AI Scientist deployment (particularly via the multi-fidelity integration gap and the current lack of validated automated novelty gates).

A Analytical Separation of Observation, Reward, and Novelty Channels

We separate three objects that are easily conflated: the *observation likelihood* $p(o \mid \theta, d)$ used for Bayesian updating, the context-dependent *validator* $V(z)$ used as a reward or acceptance signal, and any additional channel that carries evidence about novelty or significance. Observation 2 studies only the restricted special case $V(o)$ with all other context held fixed so that Berk's fixed-i.i.d. result can apply; it is not a general model of novelty or acceptance. The observations below apply standard results to fixed idealized settings. They are not new general theorems for adaptive experimental design or open-ended science.

Observation 1: posterior consistency belongs to the observation channel. In a dominated, fixed i.i.d. model $\{p_\theta : \theta \in \Theta\}$ with true distribution $p_{\theta^\star}$, posterior consistency can follow when the model identifies the truth, the prior puts positive mass on every Kullback–Leibler neighborhood of it, and suitable consistent tests separate it from complements of its neighborhoods, together with the regularity conditions needed by the relevant consistency theorem [Ghosal and van der Vaart, 2017]. Compactness, identification, and prior support stated only in a topological sense are not substitutes for those conditions. The cited result therefore does not establish consistency for the adaptive, non-stationary design sequence of an AI Scientist without an additional theorem for that regime.

Equation 1 defines a one-step information objective. Maximizing it is myopically optimal for that objective by construction, but this does not establish global optimality for a sequential scientific policy. Let $\mathcal{H}_{t-1}$ contain the past designs and observations, let d_t be measurable with respect to that history, and define $\mathrm{EIG}_t = I(\theta; O_t \mid \mathcal{H}_{t-1}, d_t)$. Posterior concentration does not require a divergent sum of these one-step values. For example, when θ is discrete with finite entropy, the conditional-information chain rule gives

$$\sum_{t\geq 1} \mathbb{E}[\mathrm{EIG}_t] = I(\theta; \mathcal{H}_\infty) \leq H(\theta), \tag{6}$$

so finite cumulative expected information is compatible with posterior concentration. If V remains outside the likelihood, it has no direct role in this fixed-model consistency statement, although policy use of V can indirectly change which future designs and observations occur.

Observation 2: a coupled validator changes the pseudo-true target only through a normalized likelihood. For a fixed design d, suppose an update uses

$$q_{\theta,d}(o) = \frac{p(o \mid \theta, d)\exp\{\beta V(o)\}}{Z_\theta(d)}, \qquad Z_\theta(d) = \int p(u \mid \theta, d)\exp\{\beta V(u)\}\, du < \infty. \tag{7}$$

Let F denote the fixed i.i.d. data-generating distribution. Under the regularity, integrability, separation, and prior-mass conditions required for a Berk-type misspecification result, posterior mass is driven toward a minimizer set rather than automatically toward a unique point:

$$\Theta^\dagger = \arg\min_{\theta\in\Theta} \mathrm{KL}(F \,\|\, q_{\theta,d}) \tag{8}$$

Only when this set is a singleton do we denote its member by $\theta^\dagger$ [Berk, 1966]. Validator-induced drift means that this minimizer set differs from the corresponding projection under $p(\cdot \mid \theta, d)$; an outcome-dependent V can alter the normalizer $Z_\theta(d)$ even when V has no explicit θ argument, but non-constancy alone neither guarantees nor is required to prove a target change. An additive constant in V cancels between numerator and normalizer, while $\beta = 0$ gives $q_{\theta,d} = p(\cdot \mid \theta, d)$. A validator used only by the policy may still alter future design selection, but that is an adaptive-design effect, not the fixed-i.i.d. conclusion of Berk [1966].

Observation 3: novelty invisible to every available channel remains unidentified. Write the latent state as $\theta = (\theta_{\text{nov}}, \psi)$ and suppose the prior factorizes as $\Pi(d\theta_{\text{nov}}, d\psi) = \Pi_{\text{nov}}(d\theta_{\text{nov}})\Pi_\psi(d\psi)$. If the joint likelihood of all observed channels depends on ψ but not on θ_{nov}, then the posterior marginal of θ_{nov} equals Π_{nov} for every dataset. This full factorization from every likelihood-relevant latent component is essential: independence only from already identified components would not exclude indirect updating through a prior-correlated nuisance variable. The novelty component is therefore not consistently identifiable from those channels. Identification requires an additional informative channel, but the premise does not imply that the channel must be human in principle. The survey instead supports the narrower empirical statement that novelty and significance judgments in the systems examined here remain human-mediated or lack a validated automated gate.

These observations support the paper's diagnostic separation of observation, reward validation, and novelty judgment. They do not prove that the empirical V-completeness ordinal is necessary for trustworthy autonomous discovery, establish causal effects of validator design, or settle the paper's open-ended and adaptive setting. The former scalar-completeness and gate-reliability/"trustworthy closure" hypotheses are withdrawn, not retained as provisional hypotheses; any future outcome study requires property-specific, gate-independent endpoints.

B Cross-System Coding Reliability

The Servo event–artifact schema is the current normative contract. It records six versioned cases from five lineages, their endpoints, artifacts, events, edges, reliability evaluations, functional relations and their per-relation claims, seven frozen source anchors, and a selection ledger. The typed functional relations are bounded by case, version, configuration, and task regime, and each carries its own per-relation support status and occurrence resolution; no single stored loop label is authoritative. The records are author-interpreted source annotations. Structural validation cannot establish semantic entailment, construct validity, or scientific outcomes.

Three independent, model-pinned LLM coders from different vendors coded the earlier 14-system rubric (13 systems had all three usable outputs). The resulting Fleiss statistics—historical `Vcalibrated` κ=0.79, loop status κ=0.74, semantic-validation presence κ=1.0, mean κ=0.68, and holistic V-completeness κ=0.39—apply only to those superseded coding fields. They are retained solely as automated audit history and are not transferred to the revised constructs.

The subsequent two-vendor recode over 14 systems produced raw agreement 13/14 and Cohen's κ=0.86 for the historical present-versus-gating distinction. That exercise documents one post-hoc construct refinement, chiefly the AI Scientist (Nature 2026) reviewer-role ambiguity; it is not a reliability study of the final faceted taxonomy. The legacy V_{present}, V_{gating}, and $V_{\text{calibrated}}$ columns therefore remain in the supplement only for reconstruction.

The frozen 42-session, three-vendor multicoder audit used a then-current manual to code 14 source records. It is retained only as a historical development audit. Its record-level token unions, Jaccard summaries, MASI-based Krippendorff α, and disagreement log neither align current events and edges nor reproduce the current functional relations and their per-relation claims. No statistic from this audit is transferred to the current contract or case statuses. Its provenance is retained in the repository revision-history record.

C Per-Relation Evidence Table

Table 7 lists every typed functional relation recorded across the six cases, with its source support, occurrence resolution, and evidence mode. It is the per-relation record behind the case-level boundary of Table 5 and the policy decomposition of Table 4.

Table 7: Relation/evidence table (Table B2): each row is one distinct (source support, occurrence resolution, evidence mode) combination carried by a typed (source role, relation type, target role, temporal scope) relation, grouped by case, so the three axes on a row correspond; a relation with several combinations spans several rows with its label repeated. Source support, occurrence resolution, and evidence mode are kept separate. Evidence mode names the kind of proposition the source supports and is not an ordinal scale of system quality or closure.

Case	Functional relation	Source support	Occurrence resolution	Evidence mode
C01	action → produces → execution (per_step)	supported	resolved	occurrence
C01	action → produces → execution (per_step)	supported	n.a.	procedure
C01	environment → produces → observation (per_step)	supported	unresolved	occurrence
C01	environment → produces → observation (per_step)	unresolved	unresolved	capability
C01	evaluation → evaluates → candidate (per_step)	supported	n.a.	procedure
C01	evaluation → evaluates → observation (per_step)	supported	resolved	occurrence
C01	evaluation → revises → artifact (cross_step)	supported	resolved	occurrence
C01	execution → reads → external (per_step)	supported	resolved	occurrence
C01	execution → reads → external (per_step)	supported	n.a.	procedure
C01	external → conditions → candidate (per_step)	supported	resolved	occurrence
C01	external → evaluates → candidate (per_step)	supported	n.a.	procedure
C01	external → selects → action (per_step)	supported	resolved	occurrence
C01	generation → produces → candidate (per_step)	supported	resolved	occurrence
C01	generation → produces → candidate (per_step)	supported	n.a.	procedure
C01	memory → conditions → policy (cross_step)	unresolved	unresolved	capability
C01	observation → updates → inquiry_state (cross_step)	supported	n.a.	procedure
C01	policy → selects → action (per_step)	supported	resolved	aggregate

continued on next page

Table B2 (continued)

Case	Functional relation	Source support	Occurrence resolution	Evidence mode
C01	policy → selects → action (per_step)	supported	n.a.	procedure
C02	evaluation → evaluates → artifact (per_step)	supported	resolved	aggregate
C02	evaluation → evaluates → artifact (per_step)	supported	n.a.	procedure
C02	evaluation → evaluates → candidate (per_step)	supported	n.a.	procedure
C02	evaluation → evaluates → candidate (per_step)	unresolved	unresolved	capability
C02	evaluation → evaluates → observation (per_step)	supported	resolved	aggregate
C02	evaluation → triggers → execution (cross_step)	supported	resolved	aggregate
C02	execution → produces → execution (per_step)	supported	resolved	occurrence
C02	execution → produces → execution (per_step)	supported	n.a.	procedure
C02	execution → reads → external (per_step)	supported	n.a.	procedure
C02	generation → produces → candidate (per_step)	supported	resolved	aggregate
C02	generation → produces → candidate (per_step)	supported	resolved	occurrence
C02	generation → produces → candidate (per_step)	supported	n.a.	procedure
C02	generation → produces → candidate (per_step)	unresolved	unresolved	capability
C02	generation → produces → candidate (per_step)	contradicted	n.a.	aggregate
C02	memory → reads → execution (per_step)	supported	n.a.	procedure
C02	memory → reads → observation (per_step)	supported	n.a.	procedure
C02	memory → writes → candidate (per_step)	supported	n.a.	procedure
C02	memory → writes → observation (per_step)	supported	n.a.	procedure
C02	observation → produces → observation (per_step)	supported	resolved	aggregate
C02	observation → produces → observation (per_step)	supported	n.a.	procedure
C03	action → produces → execution (per_step)	supported	n.a.	procedure
C03	action → reads → external (per_step)	supported	n.a.	procedure
C03	environment → produces → observation (per_step)	supported	n.a.	procedure
C03	evaluation → evaluates → artifact (per_step)	supported	n.a.	procedure
C03	evaluation → evaluates → candidate (per_step)	supported	n.a.	procedure

continued on next page

Table B2 (continued)

Case	Functional relation	Source support	Occurrence resolution	Evidence mode
C03	evaluation → evaluates → execution (per_step)	supported	n.a.	procedure
C03	evaluation → evaluates → observation (per_step)	supported	n.a.	procedure
C03	external → conditions → candidate (per_step)	supported	n.a.	procedure
C03	external → evaluates → artifact (per_step)	supported	resolved	occurrence
C03	external → evaluates → candidate (per_step)	supported	resolved	occurrence
C03	generation → produces → candidate (per_step)	supported	n.a.	procedure
C03	generation → produces → candidate (per_step)	unresolved	unresolved	capability
C03	memory → reads → observation (per_step)	supported	n.a.	procedure
C03	memory → writes → candidate (per_step)	supported	n.a.	procedure
C03	memory → writes → evaluation (per_step)	supported	n.a.	procedure
C03	memory → writes → observation (per_step)	supported	n.a.	procedure
C03	policy → selects → action (per_step)	supported	n.a.	procedure
C04	evaluation → evaluates → artifact (per_step)	supported	resolved	aggregate
C04	evaluation → evaluates → artifact (per_step)	supported	n.a.	procedure
C04	evaluation → evaluates → artifact (per_step)	unresolved	unresolved	capability
C04	evaluation → evaluates → candidate (per_step)	supported	n.a.	procedure
C04	evaluation → evaluates → candidate (per_step)	unresolved	unresolved	capability
C04	evaluation → evaluates → observation (per_step)	supported	n.a.	procedure
C04	evaluation → evaluates → observation (per_step)	unresolved	unresolved	capability
C04	execution → produces → execution (per_step)	supported	resolved	occurrence
C04	execution → produces → execution (per_step)	supported	n.a.	procedure
C04	execution → reads → external (per_step)	supported	resolved	aggregate
C04	execution → reads → external (per_step)	supported	n.a.	procedure
C04	external → conditions → candidate (cross_step)	supported	resolved	aggregate
C04	external → evaluates → artifact (per_step)	supported	resolved	aggregate
C04	external → evaluates → artifact (per_step)	supported	n.a.	procedure
C04	generation → produces → candidate (per_step)	supported	resolved	aggregate

continued on next page

Table B2 (continued)

Case	Functional relation	Source support	Occurrence resolution	Evidence mode
C04	generation → produces → candidate (per_step)	supported	n.a.	procedure
C04	generation → produces → candidate (per_step)	unresolved	unresolved	capability
C04	memory → reads → candidate (per_step)	supported	n.a.	procedure
C04	memory → writes → candidate (per_step)	supported	n.a.	procedure
C04	observation → produces → observation (per_step)	supported	n.a.	procedure
C04	policy → selects → action (per_step)	supported	n.a.	procedure
C05	action → produces → execution (per_step)	supported	resolved	occurrence
C05	action → produces → execution (per_step)	supported	n.a.	procedure
C05	environment → produces → observation (per_step)	supported	n.a.	procedure
C05	evaluation → evaluates → candidate (per_step)	supported	resolved	occurrence
C05	evaluation → evaluates → candidate (per_step)	unresolved	unresolved	capability
C05	evaluation → evaluates → observation (per_step)	supported	n.a.	procedure
C05	evaluation → updates → inquiry_state (cross_step)	supported	resolved	occurrence
C05	execution → reads → external (per_step)	supported	n.a.	procedure
C05	external → conditions → candidate (per_step)	supported	n.a.	procedure
C05	external → evaluates → candidate (per_step)	supported	resolved	occurrence
C05	generation → produces → candidate (per_step)	supported	resolved	occurrence
C05	generation → produces → candidate (per_step)	supported	n.a.	procedure
C06	action → produces → execution (per_step)	supported	resolved	aggregate
C06	action → produces → execution (per_step)	supported	n.a.	procedure
C06	environment → produces → observation (per_step)	supported	n.a.	procedure
C06	evaluation → evaluates → artifact (per_step)	supported	n.a.	procedure
C06	evaluation → evaluates → candidate (per_step)	supported	resolved	aggregate
C06	evaluation → evaluates → candidate (per_step)	supported	n.a.	procedure
C06	evaluation → evaluates → candidate (per_step)	unresolved	unresolved	capability
C06	evaluation → evaluates → observation (per_step)	supported	n.a.	procedure

continued on next page

Table B2 (continued)

Case	Functional relation	Source support	Occurrence resolution	Evidence mode
C06	`execution` → `reads` → `external` (`per_step`)	`supported`	`n.a.`	`procedure`
C06	`external` → `conditions` → `candidate` (`per_step`)	`supported`	`n.a.`	`procedure`
C06	`external` → `conditions` → `candidate` (`per_step`)	`unresolved`	`unresolved`	`capability`
C06	`external` → `evaluates` → `artifact` (`per_step`)	`supported`	`n.a.`	`procedure`
C06	`external` → `evaluates` → `candidate` (`per_step`)	`supported`	`resolved`	`aggregate`
C06	`external` → `evaluates` → `candidate` (`per_step`)	`supported`	`resolved`	`occurrence`
C06	`external` → `evaluates` → `candidate` (`per_step`)	`unresolved`	`unresolved`	`capability`
C06	`generation` → `produces` → `candidate` (`per_step`)	`supported`	`resolved`	`aggregate`
C06	`generation` → `produces` → `candidate` (`per_step`)	`supported`	`resolved`	`occurrence`
C06	`generation` → `produces` → `candidate` (`per_step`)	`supported`	`n.a.`	`procedure`
C06	`generation` → `produces` → `candidate` (`per_step`)	`unresolved`	`unresolved`	`capability`
C06	`generation` → `revises` → `candidate` (`cross_step`)	`supported`	`resolved`	`occurrence`

Legend. Per functional relation (contract section C): each row is one distinct (support status, occurrence resolution, evidence mode) combination carried by the DerivedDecisionClaims that use a relation of a given (`source_role`, `relation_type`, `target_role`, `temporal_scope`) tuple within a case, so the three axes on a row correspond; a relation with several combinations spans several rows with its label repeated. Support status, occurrence resolution and evidence mode are three separate columns and are never merged; support status uses `supported` (not `resolved`). Domains: support status ∈ {`supported`, `unresolved`, `contradicted`, `n.a.`}; occurrence resolution ∈ {`resolved`, `unresolved`, `n.a.`}; evidence mode ∈ {`architecture`, `capability`, `procedure`, `aggregate`, `occurrence`}. *Evidence mode* maps: `architecture` → structural organization; `capability` → what the system can or is designed to do; `procedure` → specified operational sequence; `aggregate` → repetition or result without event identity; `occurrence` → bounded event-level instance. Evidence modes identify the kind of proposition supported by the source and are not treated as an ordinal scale of system quality or closure.

D Per-Domain Analysis

This appendix comments on the same seven frozen anchors as Table 6. Each paragraph is bounded to one named system and source. It does not describe a domain population.

FunSearch. FunSearch [Romera-Paredes et al., 2024] records program generation with executable scoring in a bounded computational task. This anchor does not supply field-novelty evidence.

AI Feynman. AI Feynman [Udrescu and Tegmark, 2020] evaluates equation candidates against problem-local data. The anchor does not establish cross-task memory transfer.

ChemCrow. ChemCrow [Bran et al., 2024] combines tool use with reported chemical execution. Wet-lab characterization remains protocol- and artifact-specific.

GNoME. GNoME [Merchant et al., 2023] routes DFT relaxation and model-predicted stability into computational search for phase stability, selecting candidates by predicted energy under a stability threshold rather than by predictive uncertainty; its deep ensembles quantify uncertainty for generalization, not for experiment selection. DFT approximates physical energies rather than serving as a physical oracle: 736 predicted structures match ICSD entries, including 184 added after the project began, while synthesis and functional characterization remain separate [Merchant et al., 2023, Cheetham and Seshadri, 2024].

BioPlanner. BioPlanner [O'Donoghue et al., 2023] is an automatic-evaluation benchmark for LLM-generated protocol pseudocode; it does not itself generate the protocols, and protocol text alone does not establish a biological-validity feedback witness.

Manning automated social science. The selected workflow [Manning et al., 2024] uses statistical tests over generated experimental designs. This one source is not evidence about automated social science generally.

AutoML-Zero. AutoML-Zero [Real et al., 2020] routes executable benchmark results into search. Benchmark performance does not establish interpretability.

Cross-anchor reading. The rows show only that the same vocabulary can be instantiated at seven named source boundaries. They do not support aggregation, causal comparison, or domain-level generalization.